

Synchronized local completion with a fixed number of controlled moments

Complete proof; three independent mathematical reviews completed

30 September 2026

Fix $q \geq 1$ and positive constants a, b . Polynomial bounds below are in one parameter P , with exponents depending only on q, a, b and the fixed input polynomial bounds. A single common denominator, rather than separate bounds on individual denominators, is part of the conclusion.

Theorem 1. *Consider polynomially many rational polynomial probability densities w_α on $[0, 1]$, satisfying $a \leq w_\alpha \leq b$, with polynomial degrees, and let $m \geq \max(2, q)$ be polynomially bounded. Put $R = m + q + 2 + \max_\alpha \deg w_\alpha$. For each instance prescribe an integer K_α with*

$$CR^4 \leq K_\alpha \leq \text{poly}(P).$$

Suppose the numbers

$$K_\alpha \int_0^1 t^j w_\alpha(t) dt, \quad 0 \leq j \leq 2q + 1, \quad (1)$$

have one polynomial common denominator. There exist one polynomial integer L and, for every instance, rational pure densities $f_{A,\alpha}, f_{B,\alpha}$ such that:

1. Their ordered polynomial moments through degree m equal those of independent uniform and w_α variables, respectively.
2. B occupies K_α disjoint boxes, each of mass $1/K_\alpha$; A occupies the complementary gaps. The endpoints of all boxes and gaps across all instances lie on one polynomial grid.
3. On every A gap, all unnormalized moments through q are integers divided by L . On every B box, all conditional moments through q have one polynomial common denominator, shared across all boxes and all instances.
4. The numbers of cells, reciprocal lengths, reciprocal positive masses, and polynomial degrees are polynomially bounded. Conditional densities, rescaled to their cells, have fixed positive upper and lower bounds depending only on a, b, q .

No small-denominator claim is made for higher moments or coefficients.

Lemma 1 (A fixed finite set of forced quadrature nodes). *For each fixed h , one may strengthen the separated weighted quadrature lemma in `generalized_local_two_variable_completion.tex` by requiring nodes at the prescribed positions $\tau_i = i/(h+1)$, $1 \leq i \leq h$. Every sufficiently large $K \geq C_h(D+d+1)^4$ is allowed for a degree- D rule of a degree- d density with bounds a, b . All endpoint and consecutive gaps remain at least c_h/K .*

Proof. Begin with the midpoint quantile grid. For each τ_i , replace a nearest grid point by τ_i ; these chosen points are distinct once K is large. A nearest-point choice leaves distance at least half the minimum original grid spacing to every unchosen neighbor. Each move is $O_{a,b}(1/K)$. Its contribution to the normalized moment residual is $O(\text{poly}(D+d)/K^2)$, and there are only h moves. Run the same fixed-right-inverse Newton argument on the *other* nodes, holding these h

coordinates fixed. Deleting h columns from the ordinary Legendre derivative Gram changes it by at most $C_h D^2/K$; the forced moves produce another polynomially small error. The lower Gram bound survives. The correction radius is still $C_h(D+d+1)^4/K^2$, smaller than a prescribed small multiple of $1/K$. Thus the free nodes remain separated, and the fixed nodes are retained. $\square$

Proof of Theorem 1. One universal feature family. Use the cumulative-moment conditioning theorem in `../rational_designs/asymptotic/multiple_prefix_moment_conditioning.tex`. Choose the same q separated Legendre bands for every instance, with the first degree larger than $\max \deg w_\alpha + J + q + 1$, where $J = m + q + 1$. All band degrees are $O_q(R)$. With one common sufficiently small inverse-polynomial η , put

$$f_*(t) = 1 - \eta \sum_{i=1}^q P_{d_i}(2t-1), \quad M_s(t) = \int_0^t u^s f_*(u) du \quad (0 \leq s \leq q).$$

These features are within $C_q \eta$ of $t^{s+1}/(s+1)$ and have polynomially bounded derivatives. Their polynomial-multiplier family $1, M_0, \dots, M_q$, with multiplier degrees at most $J-1$, has an inverse-polynomial Gram lower bound. This holds uniformly with every weight w_α by the assumed density bounds. Legendre orthogonality gives

$$\int t^j M_s(t) w_\alpha(t) dt = \frac{1}{s+1} \int t^{j+s+1} w_\alpha(t) dt \quad (0 \leq j \leq J, 0 \leq s \leq q). \quad (2)$$

Set $h = q + 2$. Use Lemma 1 to choose, for each prescribed K_α , a separated real rule $b_{\alpha u}^0$ of sufficiently high degree $O_q(R)$. It is exact for every moment in (2) and for squares of every feature-multiplier polynomial just described. It contains all h universal control nodes $\tau_i = i/(h+1)$.

Shared rational control features and exact rank sums. Let D_0 clear the common denominators in (1). Choose one large polynomial integer L divisible by the PRODUCT $D_0(q+1)!$. At the h control indices, round the numbers $M_s(\tau_i)$ DOWN once to multiples of $1/L$, using identical values in every instance. These feature values, together with the constant feature 1, form a square $h \times h$ matrix Ψ_* . It is a small perturbation of the fixed Vandermonde-type matrix with rows $1, \tau, \tau^2/2, \dots, \tau^{q+1}/(q+1)$. Hence its inverse norm is bounded in terms of q alone. Its rational inverse has one polynomial denominator, because h is fixed and all entries have denominator dividing L .

Round the other feature values $M_s(b_{\alpha u}^0)$ to the same grid, and correct their sums to satisfy

$$\sum_u r_{s, \alpha u} = \frac{K_\alpha}{s+1} \int t^{s+1} w_\alpha(t) dt.$$

The desired sums multiplied by L are integers by (1). The control values remain fixed. After rounding down, at most K_α integer units are needed, so adding at most two units to each of the $K_\alpha - h$ free entries suffices for large enough K_α . Thus every feature changes by at most $2/L$. The same assertion holds for each of the fixed number of feature rows independently.

Hold all these rational features fixed and correct the nodes to restore every nonconstant ordinary and feature-weighted moment through J . In the antiderivative Legendre basis, the derivative evaluates $1, M_0, \dots, M_q$ times the Legendre polynomials of degrees below J . Its baseline empirical Gram is exactly the weighted continuous Gram. The conditioning theorem, polynomial derivative bounds, and the $O(\text{poly}(P)/L)$ rounding residual give the same fixed-right-inverse contraction as in the reviewed local two-variable proof. With a sufficiently large exponent for L , the node displacement is smaller than any prescribed inverse polynomial. All separations survive. The control *nodes* may move at this step; their rational *features* remain the identical columns of Ψ_* , which is what the denominator argument below requires. We now have exact ordinary moments and

$$K_\alpha^{-1} \sum_u r_{s, \alpha u} b_{\alpha u}^j = \frac{1}{s+1} \int t^{j+s+1} w_\alpha(t) dt, \quad 0 \leq j \leq J. \quad (3)$$

All feature arrays remain within $C_q\eta$ of the uniform prefix moments $b^{s+1}/(s+1)$, after choosing the rounding precision finer.

Smoothing with one common low-moment denominator. Apply the fixed- q box-smoothing argument in `../rational_designs/asymptotic/polynomial_mean_box_smoothing.tex` using the feature rows $(1, r_0, \dots, r_q)$. Choose one common inverse-polynomial halfwidth ϵ and one common polynomial center-rounding grid, sufficiently fine for all instances. Uniform bounds on all preceding condition numbers allow these choices. First solve the real uniform-box moment problem; then round centers and low conditional box moments through q on common grids, and correct the low global moment totals using exactly the control minor Ψ_* in each instance.

Here is the arithmetic point in full. The control equations have form

$$\Psi_* \mu_{\text{control},j} = K_\alpha T_{\alpha,j} - \sum_{u \notin \text{control}} \psi_{\alpha u} \mu_{\alpha u,j}.$$

Every entry on the right has one common polynomial denominator across all instances. For the sum this follows from the common feature and moment grids. For the targets it follows from (1), since ordinary targets are $K_\alpha \int t^j w_\alpha$ and feature targets are $K_\alpha \int t^{j+s+1} w_\alpha / (s+1)$, with $j \leq q$ and $j+s+1 \leq 2q+1$. The inverse of Ψ_* is the SAME rational matrix in every instance. Therefore all corrected conditional low box moments share one polynomial denominator. No least common multiple of unrelated minors is used. The remaining high moments are corrected by the degree- $(q+1)$ Legendre mode on each box; its rational coefficients may have unrestricted denominators. The reviewed argument gives positive conditional densities with fixed bounds, exact ordinary moments and exact (3). All box endpoints use the common center grid and halfwidth.

Positive completion on the gaps. Prescribe on the A gaps cumulative moments $r_{s,\alpha u}$, $0 \leq s \leq q$, before each B box, with the endpoint values 0 and $1/(s+1)$. All lie on the $1/L$ grid. On each gap realize the required moments by a polynomial of degree q , obtaining g . The elementary rescaled moment system gives

$$\|g - 1\|_{L^\infty(\text{gaps})} \leq \text{poly}_q(P)(\eta + \epsilon + \text{extnode}/\text{centererrors}),$$

while extending g by zero on the boxes adds at most $\sqrt{2K_\alpha\epsilon}$ to $\|1 - g\|_2$. The exact equations (3), and the ordinary box moments, are precisely the compatibility conditions on

$$\mathcal{S}_q = \mathcal{P}_q + \text{span}\{a^s S_t(a) : s \leq q, s+t \leq m\}.$$

Use the fixed- q quantitative smoothed quotient and positive-completion extension in `../independent_directions/quantitative_smoothed_quotient.tex`. Its losses are polynomial for fixed q and inverse-polynomial gap lengths. Choose first η sufficiently small, then ϵ , then all rounding precisions sufficiently fine. The resulting rational Riesz correction has zero moments through q on every gap and produces f_A between $1/4$ and $7/4$ there. Thus each gap moment through q is exactly a difference of entries of the $1/L$ prefix grids. The density is polynomial of degree at most m on gaps; B pieces have degree at most $q+1$. Lengths and masses are inverse-polynomial, and conditional density bounds follow by normalizing each cell.

Finally, exact integration of $\mathcal{P}_m + \text{span}\{a^r S_s(a) : r+s \leq m\}$ gives

$$\mathbb{E}[A^r B^s \mathbf{1}_{A < B}] = \frac{1}{r+1} \int b^{r+s+1} w_\alpha(b) db \quad (r+s \leq m),$$

and the ordinary and opposite-chamber moments follow as before. This proves the ordered target assertion and all denominator claims. $\square$